

**ACHARYA INSTITUTE OF TECHNOLOGY**
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**Bengaluru - 560 107****INTERNAL ASSESSMENT - II [Academic Year: 2025-26]**

Course Name:	Database Management Systems	Course Code:	BCS403
Semester/Section:	IV/ A	Max Marks:	15M
Date:	9 th June 2026	Duration:	60 Min

Note: Answer all the Questions

Sl. No.	Question	CO	BL	Marks					
1	Illustrate the normalisation process by considering the following Functional dependency and table for the following table till 3NF. Employee → Name, State code State_code → Home_state Job_code → Job_title (Employee_id, Job_code) is the candidate key.	CO3	BL3	5M					
	Employee_id				Name	Job_code	Job_title	State_code	Home_state
	E001				Alice	J01	Chef	27	Karnataka
	E001				Alice	J02	Waiter	27	Karnataka
	E002				Bob	J02	Waiter	30	Maharashtra
	E002				Bob	J03	Manager	30	Maharashtra
	E003				Alice	J01	Chef	30	Maharashtra
OR									
2	Write SQL queries by considering the given relations EMPLOYEE(Emp_ID, Name, Dept_ID, Salary) DEPARTMENT(Dept_ID, Dept_Name) 1. Find departments having more than 4 employees. 2. Find departments whose average salary exceeds ₹50,000. 3. Find employees whose name starts with 'K'. 4. Display distinct department names. 5. Count employees department-wise and sort in ascending order.	CO3	BL3	5M					
3	Explain ACID properties of database transactions.	CO4	BL2	5M					
OR									
4	Describe Triggers and Assertions in SQL with syntax and example	CO4	BL2	5M					
5	Briefly discuss the two-phase locking techniques for concurrency control.	CO5	BL2	5M					
OR									
6	Illustrate the MongoDB CRUD operations with their syntax and suitable examples.	CO5	BL2	5M					

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Course outcomes	
At the end of the course the student will be able to	
CO-1	Discuss the fundamental concepts of databases and data modeling techniques for structured data.
CO-2	Apply relational model concepts and relational algebra operations to design constrained database schemas and solve queries for real-world applications.
CO-3	Apply normalization using functional dependencies to design optimized relational schemas and SQL queries for effective database operations.
CO-4	Discuss SQL - views, assertions, triggers, and transaction processing concepts
CO-5	Discuss the fundamentals of NoSQL databases, big data storage systems and concurrency control techniques for unstructured data.
CO-6	Design and implement relational and NoSQL databases by creating database schemas, writing SQL queries, and performing basic CRUD (Create, Read, Update, Delete) operations on structured and semi-structured data.